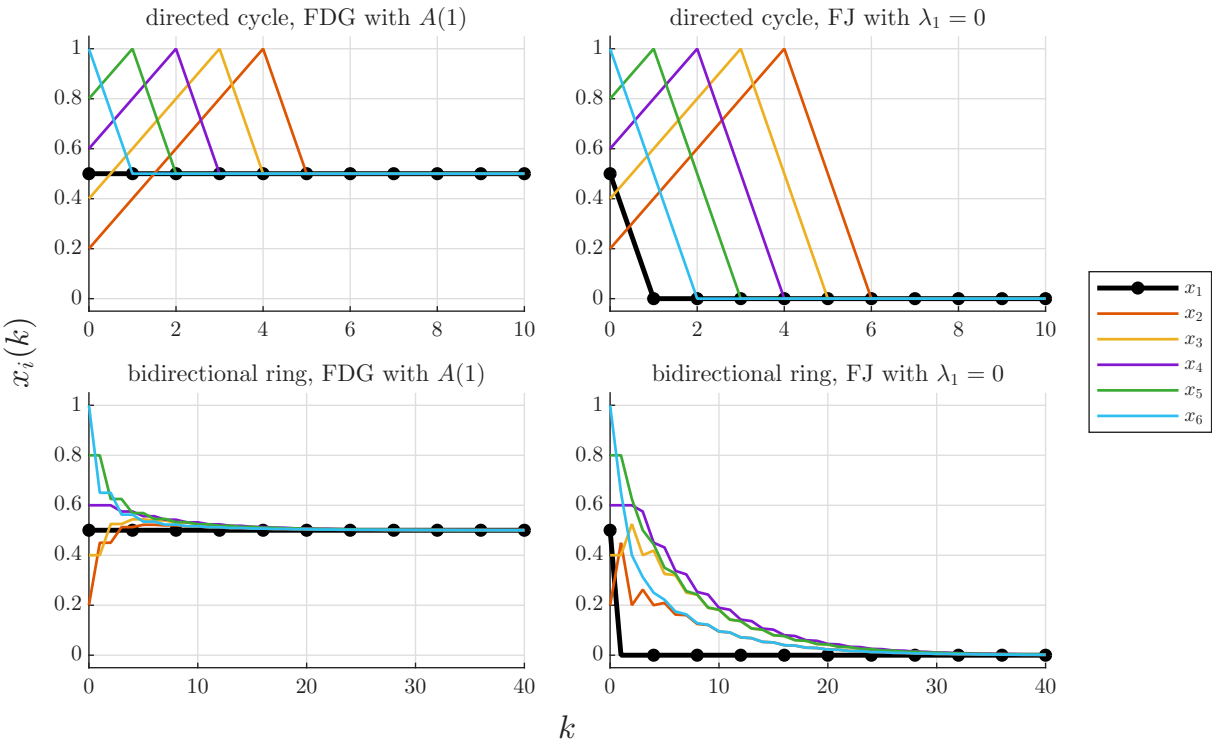


Figure 7: The models of Figure 6 started from $x_1(0) = 0.5$, with the other initial opinions and the prejudice vector $u = (0,\, 0.2,\, 0.4,\, 0.6,\, 0.8,\, 1)^\top$ unchanged, so $u_1 \neq x_1(0)$. Left: FDG with $A(1)$; right: FJ with $\lambda_1 = 0$. Top: directed cycle, $K = 10$; bottom: bidirectional ring, $K = 40$.

Figure 7 shows this difference. Moving $x_1(0)$ from 0 to 0.5 moves the stubborn agent of the FDG model, which keeps the new value forever, and the whole group follows it to 0.5. In the FJ model the same change does not touch $u_1 = 0$: agent 1 goes back to u_1 at the first step, and the group converges to the same limit Vu as before. In social terms, in the FDG model stubbornness is a memory inside the state, which is never forgotten; in the FJ model it is an external input, and the state forgets $x(0)$.

How the perturbation evolves. Figure 7 perturbs the initial state. Let $x(k)$ and $\tilde{x}(k)$ be the trajectories of Figures 6 and 7. In the FJ model the prejudice u is the same in the two runs, so their difference obeys

$$\delta(k+1) = \Lambda W\, \delta(k),$$

the same recursion as the error (10), although $e(k)$ measures the distance of one trajectory from its limit and $\delta(k)$ what is left of the perturbation. Since all agents are P-dependent, ΛW is Schur stable (Corollary 3.3): $\delta(k) \to 0$ and the two runs reach the same limit Vu, which is what asymptotic stability means. In the FDG model the same computation gives $\delta(k+1) = A\,\delta(k)$ and, by Theorem 2.1, $\delta(k) \to (w^\top \delta(0))\mathbf{1}_6 = 0.5\,\mathbf{1}_6$: the perturbation is never forgotten.

The spectra show where the difference lies. ΛW and $A(1)$ differ only in the row of agent 1, zero in ΛW and $e_1^\top$ in $A(1)$, and so do their spectra: on the bidirectional ring $\mathrm{spec}(\Lambda W) = \{0, 0, \pm\frac{1}{2}, \pm\frac{\sqrt{3}}{2}\}$ and $\mathrm{spec}(A(1)) = \{1, 0, \pm\frac{1}{2}, \pm\frac{\sqrt{3}}{2}\}$ (on the directed cycle all the eigenvalues are 0, except the eigenvalue 1 of $A(1)$). The eigenvalue 1 of $A(1)$ is that of the virtual agent $1'$ of the augmented model: in the FDG model the opinion of the stubborn agent is a state, that of $1'$, and changing $x_1(0)$ changes it; in the FJ model Figure 7 moves only the real agent 1, with $x_{1'}(0) = u_1$ unchanged, and the perturbation is forgotten.

The stubborn agent as a prejudice of its neighbours. The second rewriting removes the stubborn agent. If agent 1 is stubborn, $x_1(k) = x_1(0)$ for all k, and an agent i that listens to it with weight a_{i1} updates as

$$x_i(k+1) = \sum_{j \neq 1} a_{ij}\, x_j(k) + a_{i1}\, x_1(0),$$

which is (5) with $\lambda_i = 1-a_{i1}$, $w_{ij} = a_{ij}/\lambda_i$ and $u_i = x_1(0)$; the other agents keep $\lambda_i = 1$. The remaining $n-1$ agents obey an FJ model in which the stubborn opinion is the common prejudice of the agents that listen to it: the augmented graph read backwards, with the stubborn agent as the virtual agent. On the bidirectional ring with $t = 1$, agents 2 and 6 become prejudiced with $\lambda_2 = \lambda_6 = 1/2$ and $u_2 = u_6 = x_1(0)$, and agents $2, \dots, 6$ follow the bottom-left panel of Figure 6. Unlike in the rewriting of [5], the stubborn opinion appears only once, as a prejudice, and not also as an initial state: no initial condition of the FJ model corresponds to moving the stubborn agent alone, and the two models have the same trajectories for every initial condition.

3.2.2 Prejudices, stubbornness and convergence

A single prejudice. Figure 8 compares, with the same weight 1/2 on agent 1, the FDG model with self-weight $t = 1/2$ (left, as in the middle column of Figure 2) and the FJ model in which agent 1 is the only prejudiced agent, with attachment $1 - \lambda_1 = 1/2$ (right); the triangles mark the limits. The other two values need no new figure: for $1 - \lambda_1 = 0$ the FJ model is the FDG model of the left column of Figure 2, and for $1 - \lambda_1 = 1$ agent 1 is stubborn, as in Figure 6.

On the directed cycle the two update matrices are

$$\Lambda W = \begin{bmatrix} 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \qquad A(\tfrac{1}{2}) = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} :$$

the weight 1/2 that agent 1 takes off the agent it listens to goes, in $A(1/2)$, to the diagonal, i.e. to its current opinion, and, in the FJ model, out of the matrix, to the input $(1-\lambda_1)u_1$. The first row of ΛW sums to 1/2, so ΛW is row-substochastic (Definition 1.15).

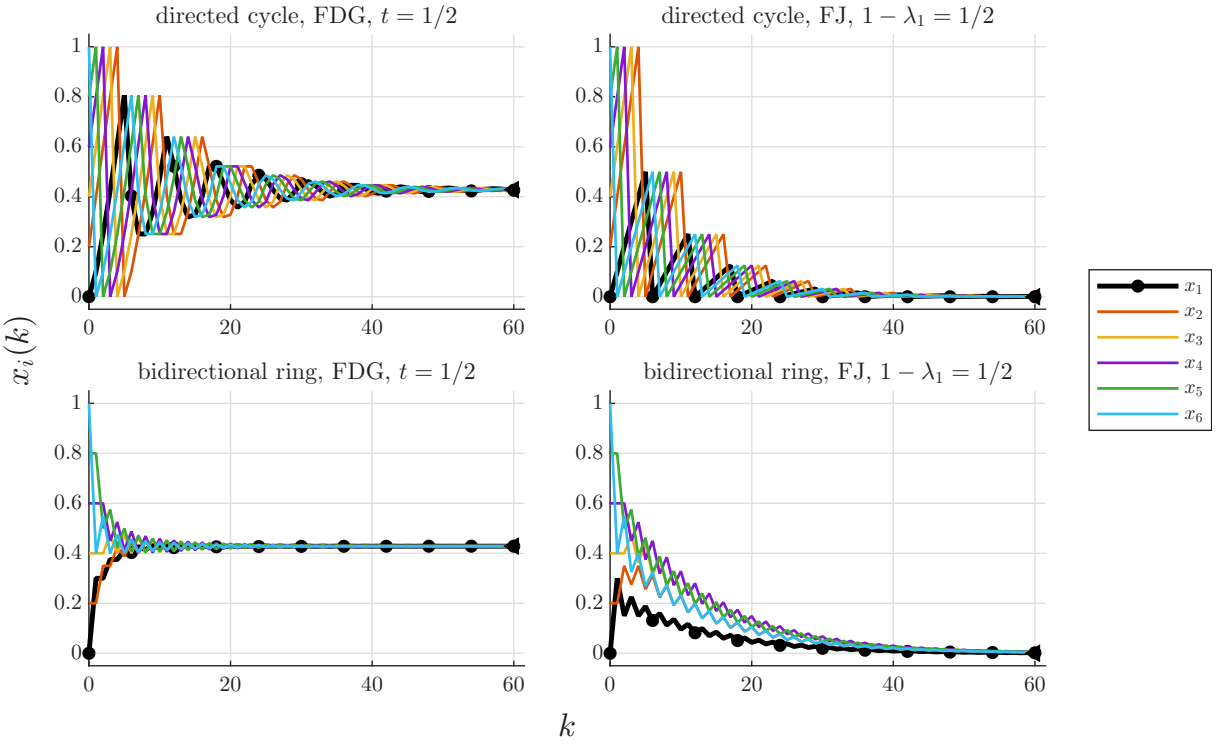


Figure 8: Opinions $x_i(k)$ with the same weight 1/2 on agent 1, with $x(0) = (0, 0.2, 0.4, 0.6, 0.8, 1)^\top$ and $K = 60$. Left: FDG model with self-weight $t = 1/2$ (matrix $A(1/2)$, current opinion); right: FJ model with $\lambda_1 = 1/2$, $\lambda_i = 1$ for $i \neq 1$ and $u = x(0)$ (attachment $1 - \lambda_1 = 1/2$, initial opinion). Top: directed cycle; bottom: bidirectional ring. The triangles mark the limits: the consensus value ≈ 0.4286 of the FDG model and $u_1 = 0$ for the FJ model.